



Explaining the Unexplained

XAI faithfulness and stability on a continuously-updating airline propensity model

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Agenda

1. Research questions and scope
2. **RQ1**, surrogate fidelity
3. **RQ2**, explanation stability across splits
4. **RQ3**, instability attribution
5. **RQ4**, cross-offer replication
6. Where the work stands, and what is left

Today

Focus is the full RQ1–RQ4 results pass. Writing the Discussion and Conclusion this week, submission **30 June**, defence **14 July**.

Four research questions

- RQ1** How accurately does the surrogate reproduce Pega's logged propensity scores on a held-out test set?
- RQ2** How consistent are feature attribution rankings across temporal, route, and culture splits?
- RQ3** When rankings shift, is the dominant source data drift (δ_d), model churn (δ_m), or explainer sensitivity (δ_e)?
- RQ4** Do the RQ1–RQ3 patterns replicate across three further offer variants with independently-updating Pega instances?

Scope

Luggage upsell, Email/Outbound channel. Primary variant: **L5B15** ($n = 64,006$). Replication: **CLUG** (generic luggage), **BookingDotCom** (accommodation), **Cartrawler** (car rental), each governed by its own Pega ADM instance.

Why a CatBoost surrogate, and the two-layer pipeline

Pega ADM uses Naive Bayes internally and is queried offline only through logged $(\mathbf{X}_i, p_i)$ tuples. We train a surrogate $g \approx f$ on those scores; SHAP and LIME then explain g :

$$f \longrightarrow g \longrightarrow \phi(\mathbf{X})$$

- **Five candidates compared** on the full fidelity suite (linear, GaussianNB, RF, LightGBM, CatBoost); CatBoost selected on $R^2 + \text{RMSE} + \tau_b$
- **TreeSHAP requires a tree model**, KernelSHAP on NB would $\sim 100\times$ compute and add its own sampling variance
- **Native categorical + missing handling**, no arbitrary ordinal encoding as a confound
- **Depth 10** selected via Breiman's 1-SD rule (CV-max at depth 12; plateau across 10–13 confirms the optimum is inside the grid)

RQ1, surrogate fidelity across all four variants

Variant	R^2	Spearman ρ	Kendall τ_b	RMSE	D_{KS}
L5B15	0.927	0.933	0.785	0.0026	0.087
CLUG	0.954	0.920	0.760	0.0026	0.082
BookingDotCom	0.793	0.949	0.811	0.0002	0.064
Cartrawler	0.912	0.615	0.461	0.0005	0.156

Reading

Pointwise fit is strong on three of four variants. Cartrawler's low rank fidelity ($\rho = 0.615$) is carried forward as a caveat on its downstream attribution numbers. Moderate D_{KS} across the board is expected: depth was selected on ρ_S , not a distributional criterion, and rank order is the operationally relevant property for Pega's scoring.

RQ2, stability matrix on L5B15

Spearman ρ_S on the full feature ranking across the five L5B15 splits:

Split	SHAP			LIME			Decision Tree		
	ρ_S	J_5	J_{10}	ρ_S	J_5	J_{10}	ρ_S	J_5	J_{10}
Temporal (early vs late)	0.996	1.000	1.000	0.989	1.000	0.818	0.855	0.667	0.667
Route 1: ORY→NCE vs ORY→OPO	0.994	1.000	1.000	0.986	0.667	0.818	0.510	0.429	0.538
Route 2: ORY→OPO vs NCE→ORY	0.990	1.000	0.818	0.858	1.000	0.818	0.584	0.250	0.667
Route 3: NCE→ORY vs TLS→ORY	0.987	0.667	0.818	0.992	1.000	1.000	0.817	0.667	0.667
Culture (fr-FR vs nl-NL)	0.958	0.667	1.000	0.949	1.000	0.667	0.666	0.429	0.667

Pattern

SHAP very stable everywhere ($\rho_S \geq 0.958$). **LIME tracks SHAP closely**, with one dip on route pair 2. **DT collapses** on route and culture splits (ρ_S down to 0.510). Bootstrap validation (10 re-fits, same data) confirms the DT gap on temporal, route 1, route 2, and culture is real signal, not fitting noise. Route 3 falls inside the bootstrap noise floor and is not interpreted as instability.

RQ3, decomposing instability into its sources

We decompose $\pi = 1 - \rho_S$ into three sources, monitored separately:

δ_d , **data drift**

KS on propensity. Flagged if ≥ 0.10 .

δ_m , **model churn**

Jaccard on active modelVersion IDs. Flagged if ≤ 0.10 .

δ_e , **explainer gap**

$\rho_{\text{SHAP}} - \rho_{\text{DT}}$ (refit),
 $\rho_{\text{SHAP}} - \rho_{\text{LIME}}$ (sampling).
Descriptive.

Split	δ_d	δ_m	δ_e^{DT}	δ_e^{LIME}	Source
L5B15 temporal	0.356	0.002	0.141	0.007	Both
Route 1 (ORY→NCE vs ORY→OPO)	0.099	0.481	0.484	0.008	Explainer
Route 2 (ORY→OPO vs NCE→ORY)	0.094	0.547	0.406	0.132	Explainer
Route 3 (NCE→ORY vs TLS→ORY)	0.086	0.525	0.171	-0.005	Explainer
Culture (fr-FR vs nl-NL)	0.081	0.595	0.291	0.009	Explainer

Headline finding (RQ3)

On the same-window subgroup splits neither the data nor the model snapshots change, yet DT rankings differ substantially across them. **Under controlled conditions, the dominant source of observed instability is the explanation method itself.** SHAP is the most stable; LIME tracks SHAP closely (sampling variance is small).

RQ4 (i), temporal and route-pair replication

Variant	δ_d	δ_m	δ_e^{DT}	δ_e^{LIME}	Source
<i>Temporal splits</i>					
CLUG	0.390	0.002	0.421	0.005	Both
BookingDotCom	0.246	0.002	0.166	0.006	Both
Cartrawler	0.721	0.002	0.373	0.025	Both
<i>Route pair 1 (ORY→NCE vs ORY→OPO)</i>					
CLUG	0.052	0.406	0.363	0.015	Explainer
BookingDotCom	0.116	0.416	0.711	−0.036	Dist. shift
Cartrawler	0.079	0.404	0.512	0.009	Explainer
<i>Route pair 2 (ORY→OPO vs NCE→ORY)</i>					
CLUG	0.116	0.404	0.266	0.197	Dist. shift
BookingDotCom	0.224	0.352	0.471	−0.024	Dist. shift
Cartrawler	0.059	0.361	0.757	0.367	Explainer
<i>Route pair 3 (NCE→ORY vs TLS→ORY)</i>					
CLUG	0.069	0.426	0.490	0.015	Explainer
BookingDotCom	0.062	0.396	0.333	−0.010	Explainer
Cartrawler	0.062	0.376	0.571	0.010	Explainer

Temporal replicates fully (all → Both). **Routes** mostly replicate the L5B15 Explainer pattern; rows that flip to Dist. shift on borderline δ_d still carry the larger numerical contribution from δ_e^{DT} (0.27–0.76).

RQ4 (ii), culture split, the one place L5B15 does not replicate

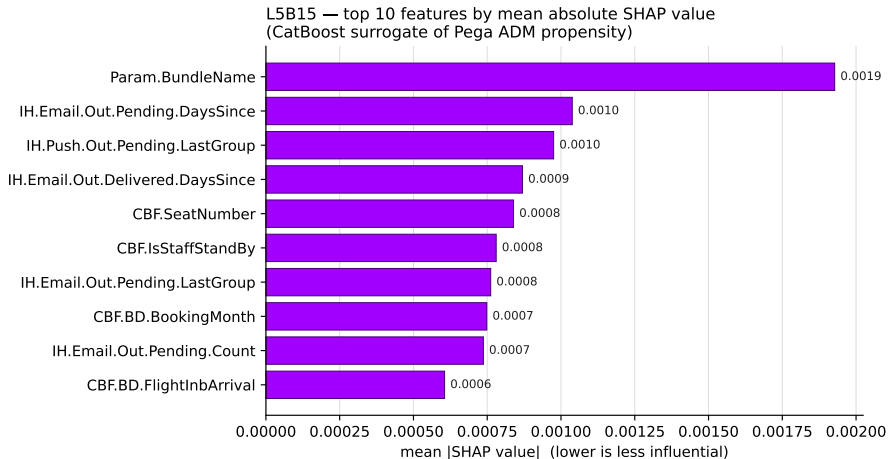
Variant	δ_d	δ_m	δ_e^{DT}	δ_e^{LIME}	Source
L5B15 (primary)	0.081	0.595	0.291	0.009	Explainer
CLUG	0.114	0.464	0.383	-0.017	Dist. shift
BookingDotCom	0.718	0.529	0.054	-0.060	Dist. shift
Cartrawler	0.689	0.564	0.177	0.352	Dist. shift

Interpretation

L5B15's culture-driven propensity shift sits just below the threshold ($\delta_d = 0.081$); the three replication variants all cross it, with BookingDotCom and Cartrawler showing very large shifts ($\delta_d > 0.68$). Customer culture (FR vs NL) maps onto materially different propensity distributions for the cross-sell and third-party offers but not for the luggage upsell. **The methodological protocol generalises; the substantive role of any single feature dimension is product-context-specific.**

What SHAP says are the top features on L5B15

Top 10 features by mean absolute SHAP value, CatBoost surrogate:



Where the work stands, and what is left

Done

- Data, methodology, RQ1–RQ4 results chapters written; tables and figures finalised
- Three-way instability attribution operationalised and replicated across four variants
- Sensitivity analyses in appendix: PSI bin count, Jaccard threshold, DT bootstrap, CV depth

This week (1–7 June)

- Write the **Discussion** (link findings to the wider XAI literature, articulate limitations beyond what is already documented in Section 2.6)
- Write the **Conclusion** (summary of contributions + future directions)

Timeline to defence

- 8–29 June, finalisation: full read-through, supervisor feedback round, polish
- **30 June** submission • **14 July** defence

Open questions for the supervisors

1. **Pega NB as a “gold-label” ranking?** Pega ADM exposes per-predictor importance directly via univariate NB likelihood contributions and bin lifts (Health Check predictor overview). Should we incorporate this as a reference ranking to validate the surrogate-based attributions, or stay strictly with the true black-box framing the methodology is built on?
2. **Discussion scope:** how broadly should the discussion engage with the wider XAI literature given the single-airline setting, beyond the limitations already in Section 2.6?
3. **Conclusion contributions:** keep the two-level framing (methodological protocol + empirical evidence under continuous online updating), or restructure?

Thank you

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